

eCH-0271 to DCAT-AP CH Mapping Documentation

Overview

This document describes the mapping from eCH-0271 (Swiss geocat.ch metadata standard) to DCAT-AP CH (Data Catalog Vocabulary - Application Profile for Switzerland).

The transformation is implemented in the XSLT stylesheet `dcat-ap-ch-core-dataset.xsl` and produces RDF/XML output conforming to the [DCAT-AP CH specification](#).

Key Features

- **Multilingual support:** Handles all four Swiss national languages (DE, FR, IT, EN) plus Romansh (RM)
- **Organization mapping:** Uses `organization-mapping.xml` to map geocat.ch organizations to `opendata.swiss` slugs
- **Intelligent slug generation:** Automatically generates publisher slugs from organization acronyms or names
- **License mapping:** Converts ISO constraints to DCAT-AP CH license URIs (`terms_open`, `terms_by`, `terms_ask`, etc.)
- **Format detection:** Automatically detects distribution formats from protocols and file extensions
- **EU vocabulary alignment:** Maps ISO code lists to EU Named Authority Lists

Dataset Level Mapping

-> See mapping-eCH0271-dcat-ap-ch table (tab `dcat_ap_ch_mapping`)

Distribution Level Mapping

Distributions are created for online resources with specific protocols:

- `WWW:DOWNLOAD*` - Download services
- `OGC:WMTS` - WMTS tile services
- `OGC:WFS` - WFS feature services
- `OGC:WMS` - WMS map services
- `ESRI:REST` - ArcGIS REST services
- `LINKED:DATA` - Linked data services
- `MAP:Preview` - Map preview pages

-> See mapping-eCH0271-dcat-ap-ch table (tab `dcat_ap_ch_distribution_mapping`)

Code Mappings

ISO Topic Category to EU Data Theme

ISO Topic Category	EU Data Theme	URI
imageryBaseMapsEarthCover, location, elevation, boundaries, planningCadastre, geoscientificInformation, structure	REGI (Regions and cities)	http://publications.europa.eu/resource/authority/data-theme/REGI

environment, biota, oceans, inlandWaters, climatologyMeteorologyAtmosphere	ENVI (Environment)	http://publications.europa.eu/resource/authority/data-theme/ENVI
society	SOCI (Population and society)	http://publications.europa.eu/resource/authority/data-theme/SOCI
health	HEAL (Health)	http://publications.europa.eu/resource/authority/data-theme/HEAL
transportation	TRAN (Transport)	http://publications.europa.eu/resource/authority/data-theme/TRAN
farming	AGRI (Agriculture)	http://publications.europa.eu/resource/authority/data-theme/AGRI
economy	ECON (Economy and finance)	http://publications.europa.eu/resource/authority/data-theme/ECON

ISO Maintenance Frequency to EU Frequency

ISO Frequency Code	EU Frequency Code	Label
continual	CONT	Continuous
daily	DAILY	Daily
weekly	WEEKLY	Weekly
fortnightly	BIWEEKLY	Biweekly
monthly	MONTHLY	Monthly
quarterly	QUARTERLY	Quarterly
biannually	ANNUAL_2	Semiannual
annually	ANNUAL	Annual
asNeeded, irregular	IRREG	Irregular
(empty/unknown)	UNKNOWN	Unknown

ISO 639 Language Codes

ISO 639-2 (3-letter)	ISO 639-1 (2-letter)	EU Code
ger, deu	de	DEU
fre, fra	fr	FRA
ita	it	ITA
eng	en	ENG
roh	rm	ROH

Distribution Format Mapping

Protocol/Extension	Format Code	URI
OGC:WMS*	WMS_SRVC	http://publications.europa.eu/resource/authority/file-type/WMS_SRVC

OGC:WMTS*	WMTS_SRVC	http://publications.europa.eu/resource/authority/file-type/WMTS_SRVC
OGC:WFS*	WFS_SRVC	http://publications.europa.eu/resource/authority/file-type/WFS_SRVC
ESRI:REST*	REST	http://publications.europa.eu/resource/authority/file-type/REST
MAP:Preview*	HTML	http://publications.europa.eu/resource/authority/file-type/HTML
*.shp	SHP	http://publications.europa.eu/resource/authority/file-type/SHP
*.gpkg	GPKG	http://publications.europa.eu/resource/authority/file-type/GPKG
*.geojson	GEOJSON	http://publications.europa.eu/resource/authority/file-type/GEOJSON
*.json	JSON	http://publications.europa.eu/resource/authority/file-type/JSON
*.gml	GML	http://publications.europa.eu/resource/authority/file-type/GML
*.kml	KML	http://publications.europa.eu/resource/authority/file-type/KML
*.csv	CSV	http://publications.europa.eu/resource/authority/file-type/CSV
*.xml	XML	http://publications.europa.eu/resource/authority/file-type/XML
*.zip	ZIP	http://publications.europa.eu/resource/authority/file-type/ZIP
*.pdf	PDF	http://publications.europa.eu/resource/authority/file-type/PDF
*.html, *.htm	HTML	http://publications.europa.eu/resource/authority/file-type/HTML
(unknown)	UNSPECIFIED	http://publications.europa.eu/resource/authority/file-type/UNSPECIFIED

License Mapping

Constraints text in `mco:useLimitation` or `mco:otherConstraints` is analyzed to determine the license:

Constraint Text Contains	License URI
"Opendata BY-ASK"	http://dcat-ap.ch/vocabulary/licenses/terms_by_ask
"Opendata BY"	http://dcat-ap.ch/vocabulary/licenses/terms_by
"Opendata ASK"	http://dcat-ap.ch/vocabulary/licenses/terms_ask
"Opendata OPEN", "Utilisation libre", "Freie Nutzung"	http://dcat-ap.ch/vocabulary/licenses/terms_open
"CC0"	https://creativecommons.org/publicdomain/zero/1.0/
"CC BY 4.0"	https://creativecommons.org/licenses/by/4.0/
(default fallback)	http://dcat-ap.ch/vocabulary/licenses/terms_open

Organization Publisher Mapping

The transformation uses a configurable organization-mapping.xml file to map [geocat.ch](https://www.geocat.ch) organization names to opendata.swiss publisher slugs. This ensures consistent publisher identifiers across both platforms.

Mapping Priority

1. **Exact match:** [geocat.ch](https://www.geocat.ch) organization name matches @geocatName in mapping file
2. **Base name exact match:** Organization name before comma matches @geocatName
3. **Contains match:** @geocatName contains the organization name
4. **Acronym fallback:** Use organization acronym from che:organisationAcronym (slugified)
5. **Name fallback:** Use organization name (slugified)

Slug Generation

The `local:slugify()` function normalizes organization names:

- German umlauts: ü→u, ö→o, ä→a, ß→ss
- French accents: é/è/ê→e, à/â→a, ç→c
- Convert to lowercase
- Replace non-alphanumeric characters with hyphens
- Remove leading/trailing hyphens and consecutive hyphens

Example: "Bundesamt für Umwelt (BAFU)" → "bafu" or "bundesamt-fur-umwelt"

URI Patterns

Dataset URI

```
https://ckan.opendata.swiss/perma/{uuid}@{publisher-slug}
```

Example:

```
https://ckan.opendata.swiss/perma/550e8400-e29b-41d4-a716-446655440000@bafu
```

Publisher URI

```
https://opendata.swiss/organization/{publisher-slug}
```

Example:

```
https://opendata.swiss/organization/bafu
```

[geocat.ch](https://www.geocat.ch) Relation URI

```
https://www.geocat.ch/geonetwork/srv/ger/catalog.search#/metadata/{uuid}
```

Example:

```
https://www.geocat.ch/geonetwork/srv/ger/catalog.search#/metadata/550e8400-e29b-41d4-a716-446655440000
```

Multilingual Handling

All text fields (title, description, keywords) support multilingual content:

1. **Default language:** Extracted from `gco:CharacterString` with language from `mdb:defaultLocale/*/lan:language`
2. **Alternative languages:** Extracted from `lan:PT_FreeText/lan:textGroup/lan:LocalisedCharacterString/@locale`

The transformation:

- Resolves locale IDs to language codes via `mdb:locale` or `mdb:otherLocale`
- Converts ISO 639-2 (3-letter) codes to ISO 639-1 (2-letter) for `xml:lang` attributes
- Skips unsupported or unmapped language codes (returns empty string instead of 'und')

Example output:

```
< dct:title xml:lang="de">Deutscher Titel</ dct:title>
< dct:title xml:lang="fr">Titre français</ dct:title>
< dct:title xml:lang="it">Titolo italiano</ dct:title>
< dct:title xml:lang="en">English title</ dct:title>
```

Date Handling

All dates are formatted as ISO 8601 DateTime with timezone:

- Input formats accepted: YYYY-MM-DD or YYYY-MM-DDTHH:MM:SS (with or without timezone)
- Output format: YYYY-MM-DDTHH:MM:SS+00:00
- Timezone 'Z' is converted to '+00:00' for better DCAT-AP CH compatibility

Implementation Notes

XPath Selection Priority

For contact/publisher organization selection:

1. `pointOfContact` with role "publisher"
2. `pointOfContact` with role "owner"
3. `pointOfContact` with role "pointOfContact"
4. First metadata contact

Distribution Filtering

Only online resources with specific protocol prefixes are converted to distributions:

- `WWW:DOWNLOAD*` - downloadable files
- `OGC:WMTS`, `OGC:WFS`, `OGC:WMS` - OGC web services
- `ESRI:REST` - ArcGIS services
- `LINKED:DATA` - linked data endpoints
- `MAP:Preview` - map preview pages

Resources with `WWW:LINK*` are used for landing pages, not distributions.

Mandatory Fields Handling

The transformation ensures all mandatory DCAT-AP CH properties are present:

- If no issued date exists, creation date is used as fallback
- If no license is found, defaults to `terms_open`
- If no format can be determined, uses `UNSPECIFIED`

- Contact point requires at least email address to be included

References

- [DCAT-AP CH Specification](#)
- [ISO 19115-3 Standard](#)
- [EU Named Authority Lists](#)
- [geocat.ch Metadata Catalog](#)
- [opendata.swiss Open Data Portal](#)

Version History

- **Current version:** Based on XSLT stylesheet in iso19115-3.2018.ch plugin
- **Target DCAT version:** DCAT 2.0 / DCAT-AP CH
- **Last updated:** February 2026